

Algorithm 6: BIC-cherry + multilayered expansion (weak)

Require: coloured digraph (G, σ) on L , maximal number of layers K (default $K = |L|$)

Ensure: leaf-coloured network (N, σ)

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1:  $V(N) \leftarrow L \cup \{\rho\}$ ;  $E(N) \leftarrow \emptyset$ ;  $W \leftarrow \emptyset$ 
2: if  $|L| = 2$  then add edges  $(\rho, x)$  for all  $x \in L$  and return  $(N, \sigma)$ 
3: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ BIC-cherry network
4:   add  $p_{xy}$  with edges  $(\rho, p_{xy}), (p_{xy}, x), (p_{xy}, y)$ 
5: end for
6: for all  $\{x, y\} \subseteq L$  with  $\sigma(x) \neq \sigma(y)$  do ▷ expansion, layer 1
7:   for all  $(a, b) \in ((x, y), (y, x))$  with  $(a, b) \notin E(G)$  do
8:      $(b', r) \leftarrow \text{FINDCANDIDATE}(G, a, b)$ 
9:     add new vertex  $q$  with edges  $(p_{xy}, q), (q, a), (q, b')$  ▷  $[ab : ab']$ 
10:    remove edge  $(p_{xy}, a)$  ▷ shortcut
11:    if not  $r$  then  $W \leftarrow W \cup \{(b', a, q)\}$  ▷ pending correction
12:  end for
13: end for
14:  $\ell \leftarrow 2$ 
15: while  $W \neq \emptyset$  and  $\ell \leq K$  do ▷ layers 2, 3, ...
16:    $W' \leftarrow \emptyset$ 
17:   for all  $(a, b, u) \in W$  do
18:      $(b', r) \leftarrow \text{FINDCANDIDATE}(G, a, b)$ 
19:     add new vertex  $q$  with edges  $(u, q), (q, a), (q, b')$ 
20:     remove edge  $(u, a)$ 
21:     if not  $r$  then  $W' \leftarrow W' \cup \{(b', a, q)\}$ 
22:   end for
23:    $W \leftarrow W'$ ;  $\ell \leftarrow \ell + 1$ 
24: end while
25: return  $(N, \sigma)$ 
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26: function FINDCANDIDATE( $G, a, b$ )
27:    $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G), (v, a) \in E(G)\}$ 
28:   if  $C \neq \emptyset$  then return (random element of  $C$ , true)
29:    $C \leftarrow \{v \in L : \sigma(v) = \sigma(b), (a, v) \in E(G)\}$ 
30:   if  $C = \emptyset$  then error ▷  $(G, \sigma)$  not sink-free
31:    $C^* \leftarrow \{v \in C : \exists w \in L \text{ with } \sigma(w) = \sigma(a), (v, w) \in E(G), (w, v) \in E(G)\}$ 
32:   if  $C^* \neq \emptyset$  then  $C \leftarrow C^*$ 
33:   return (random element of  $C$ , false)
34: end function
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